

Acceptance report - crate 4471-B

Forty humidity loggers, rev C, received 14 March. This report covers the incoming check: enclosure, power rail, and a four-hour soak against the reference meter in bay 2.

Packing was intact and the desiccant sachets were still blue. Two units had scuffed lids, cosmetic only, and are noted here so that they are not raised again at the outgoing check.

One unit, serial 4471-B-17, would not enumerate over USB until the cable was reseated. It has been kept in the batch but is flagged on page 3.

All firmware reported v1.6.5. No unit needed reflashing.

Measurements

Soak conditions: 22.4 C, 45 %RH nominal, four hours, chamber door shut throughout. Readings sampled every 30 seconds and averaged per minute.

Drift over the soak, worst unit	1.1 %RH
Drift over the soak, median	0.4 %RH
Settling time after power-up	3 min 50 s
Rail under radio transmit	3.11 V
Units outside the 2 %RH band	0 of 40

The settling figure is the reason the logger discards its first 240 seconds. Nothing in this batch settled slower than that, so the discard window can stay where it is for now.

Findings and next steps

1. Serial 4471-B-17 enumerated only after the cable was reseated. Retest with a known good cable before it ships. If it repeats, the connector is the suspect, not the board.
2. The 3.3 V rail sags to 3.11 V while the radio transmits. No reset was seen in four hours, but the margin is thinner than the schematic suggests and should be measured again at low temperature.
3. Nothing in the log tells a genuine flat line from a stuck bus. A heartbeat counter is proposed for the next revision.

Batch accepted. Recheck 4471-B-17 before dispatch.